

Jayraj Kiran Ladkat

✉ jayrajkladkat@gmail.com | 📞 +91 97170 61462 | 📍 Gurugram / Pune, India | 🌐 jayrajkl.com | 🌐 [jayraj-kl](https://jayraj-kl.github.io) | 📄 [jayraj-ladkat](https://jayraj-ladkat.github.io)

EDUCATION

B.Tech. in Computer Science and Engineering

10/2022 – 10/2026

MIT World Peace University, Pune, Maharashtra

Relevant Coursework: Data Structures & Algorithms, Distributed Systems, DBMS, Operating Systems, Computer Networks, OOP

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C/C++, Go (Basic)

Backend & Systems: Node.js, FastAPI, Flask, PostgreSQL, Apache Kafka, NATS, Redis, REST APIs, WebSockets

AI & ML Systems: Agentic RAG, LangGraph, LangChain, PyTorch, Mixed-Precision QAT, Pinecone

Cloud & DevOps: Docker, AWS (ECS, S3), CI/CD Automation, Daytona Sandboxes, Linux, Git, Nginx, Prisma

EXPERIENCE

TIGER-Lab, University of Waterloo

Waterloo, Canada (Remote)

Software Engineering Intern (Research)

07/2026 – Present

- Engineered an automated multi-step generation pipeline (script → code → render → narration) for dynamic educational videos, reducing end-to-end animation synthesis latency by 45%.
- Architected an evolutionary prompt optimization engine to dynamically calibrate multi-agent LLM prompts, boosting pedagogical clarity and instructional alignment scores by 30%.

Publicis Sapient

Gurugram, Haryana, India

Software Engineering Intern

08/2025 – 12/2025

- Optimized an AI-agent synthetic data-generation pipeline, slashing token consumption by 35% per run via prompt compression, structured caching, and schema enforcement.
- Enhanced an AI talent evaluation platform with client-side request prefetching and asynchronous assessment workers, cutting recruiter review turnaround time by 40%.

Elixir AI

Pune, Maharashtra, India

Software Engineering Intern

06/2025 – 08/2025

- Developed full-stack features for the *Climate Smart Public Health Platform* (Harvard University & Malagasy Ministry of Health collaboration) using Flask, React, and PostgreSQL/PostGIS.
- Optimized spatial PostGIS queries across large-scale geospatial raster/vector datasets and containerized services using Docker, ensuring reproducible deployments presented to international stakeholders.

Centre for Development of Advanced Computing (C-DAC)

Pune, Maharashtra, India

Cloud & Full-Stack Intern

01/2025 – 02/2025

- Integrated NATS pub/sub asynchronous messaging into microservice pipelines, eliminating synchronous HTTP blocking and decreasing peak-load request response times by 30%.

SELECTED PROJECTS

Dexterio — Autonomous Educational Animation Generation Engine

2026

Python, TypeScript, Apache Kafka, Docker, Daytona, LangGraph, PostgreSQL

[\[Code\]](#)

- Engineered an Agentic RAG engine orchestrating multi-step storyboard planning and modular Manim animation code generation with PostgreSQL-backed LangGraph checkpoints for fault-tolerant thread resumption.
- Architected a distributed asynchronous worker fleet using Apache Kafka and isolated Daytona/Docker VM sandboxes to securely compile, validate, and render untrusted generated Python code.

One Stop Deployment — Automated Microservice CI/CD Platform

2024

TypeScript, Docker, AWS (ECS, S3), Apache Kafka, Reverse Proxy, Node.js

[\[Code\]](#)

- Developed a distributed one-click deployment engine automating build and containerization workflows for GitHub repositories using containerised Docker build workers and AWS ECS/S3.
- Implemented real-time build and deployment log streaming via Apache Kafka and designed a dynamic reverse-proxy for wildcard subdomain routing to deliver instant application preview URLs.

Demex — Distributed Full-Stack Search Engine

2024

TypeScript, Next.js, PostgreSQL, Node.js

[\[Code\]](#)

- Built a high-throughput, robots.txt-compliant web crawler in TypeScript that recursively indexes web graph topologies into PostgreSQL using optimized relational graph schemas.
- Implemented a ranking pipeline combining TF-IDF lexical matching and PageRank link analysis; built a Next.js search UI supporting real-time query completion and sub-100ms result previews.

SELECTED PUBLICATIONS

- Co-Adaptation over Allocation: KL-Guided Mixed-Precision Quantization-Aware Training** – *IEEE HiPC 2026* [\[Accepted Poster\]](#)

- Real-World Scalability of PostgreSQL: Practical Techniques Use Case Study** – *IEEE ICBDS 2025* [\[IEEE Xplore 11377486\]](#)